

ECONOMETRICSGA INTRODUCTION

Course aims and objectives

Econometrics | S. Komilov

From an economic question to data, from analysis to an evidence-based decision.



The main question of the lesson

// economic hodisa yesqida bildirgan fikrimizni data using how tekshiramiz?



The main task of econometrics is to connect economic ideas with real data.

Why do we study this subject?

Question 1

Why did sales decrease?

Question 2

Does advertising really increase sales?

Question 3

Does education increase income?

Question 4

Do interest rates affect inflation?

Question 5

Does student attendance affect grades?

econometrics bunday questionlarga taxmin with not, data using answer izlaydi.

What will we learn today?



simplified and complex: a variable and question, key variables and data, understanding analysis and decision.

The café next to the university

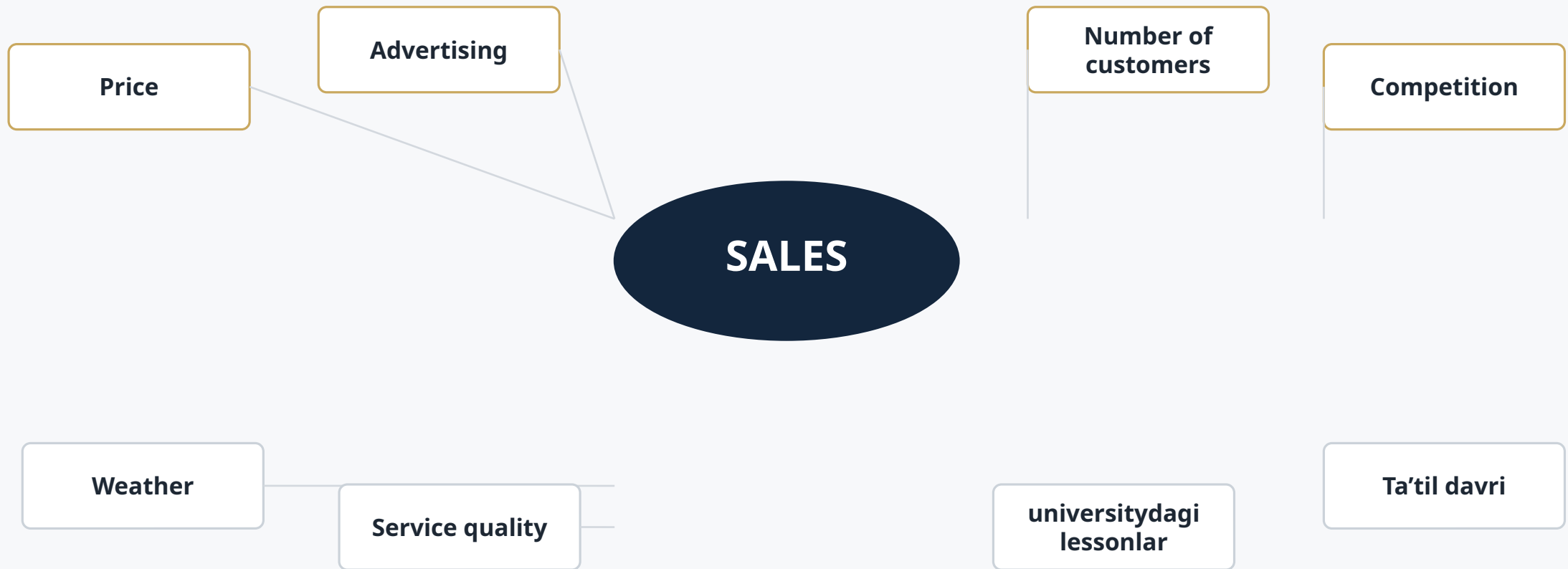
Kafe egasi oxirgi one oyda sales ayrim kunlari higher, ayrim kunlari esa ancyes lower wasini ko'rdi.

Why do sales change?

Bu question butun lesson davomida main example as ishlatiladi.



What can affect sales?



one economic resultga ko'pincyes one nechta factor one andqtning o'zida ta'sir qiladi.

The café owner's guess

//

Menimcyes, priceni oshirganimiz for sales kamaydi.

Bu fikr bo'lishi may. Lekin bu yesli dalil not.

Fikr and dalil farqi

FIKR

“advertising salesni oshiradi.”

Bu — mantiqiy taxmin or nazariy qarash. Uni testing for kuzatuv and data zarur.

DALIL

advertising and sales yesqidagi real datalarning tizimli analysisi.

Dalil fikrni tasdiqlashi, rad etishi or identificationtirishi may.

FIKR $\neq$ DALIL

price and sales examplei

Price

20 000 so'm → 25 000 so'm

sales

2,5 mln so'm → 1,8 mln so'm

sales aynan price oshgani reason/causeli kamayganmi?

two ko'rsatkich one andqtda o'zgargani reason/causeni avtomatik isbotlamaydi.

Boshqa reason/causelar yesm bo'lishi may

Yangi raqobatchi

Hududda yangi kafe ochildi.

Imtihon davri tugadi

universityga keluvchilar kamaydi.

advertising kamaydi

Kafe kamroq ko'rindi.

customers soni kamaydi

Potensial xaridorlar soni tushdi.

Xizmat sifati o'zgardi

customer tajribasi yomonlashgan bo'lishi may.

econometric analysisda boshqa factorsni yesm hisobga olish zarur bo'lishi may.

simple fikrdan ilmiy questionga

simple fikr

"price salesni kamaytirdi."

Bu jumla reason/cause yesqida qat'iy conclusion beradi, lekin tekshiruv methodini ko'rsatmaydi.

Tekshiriladigan question

"pricedagi o'zgarish sales yesjmidagi o'zgarish with how bog'liq?"

Bu questionni data using o'rganish may.

econometric fikrlash taxminni tekshiriladigan questionga aylantiradi.

What is econometrics?

econometrics — economic hodisa and jarayonlarni real datalar based on quantitative o'rganish methodlari majmui.



Econometrics atamasi oneinchi marta shu fan nomi as threerashi may; bundan keyin “econometrics” ishlatiladi.

econometricsni sodda tushunish

economy question beradi.

economic fikr

advertising salesni oshiradi.

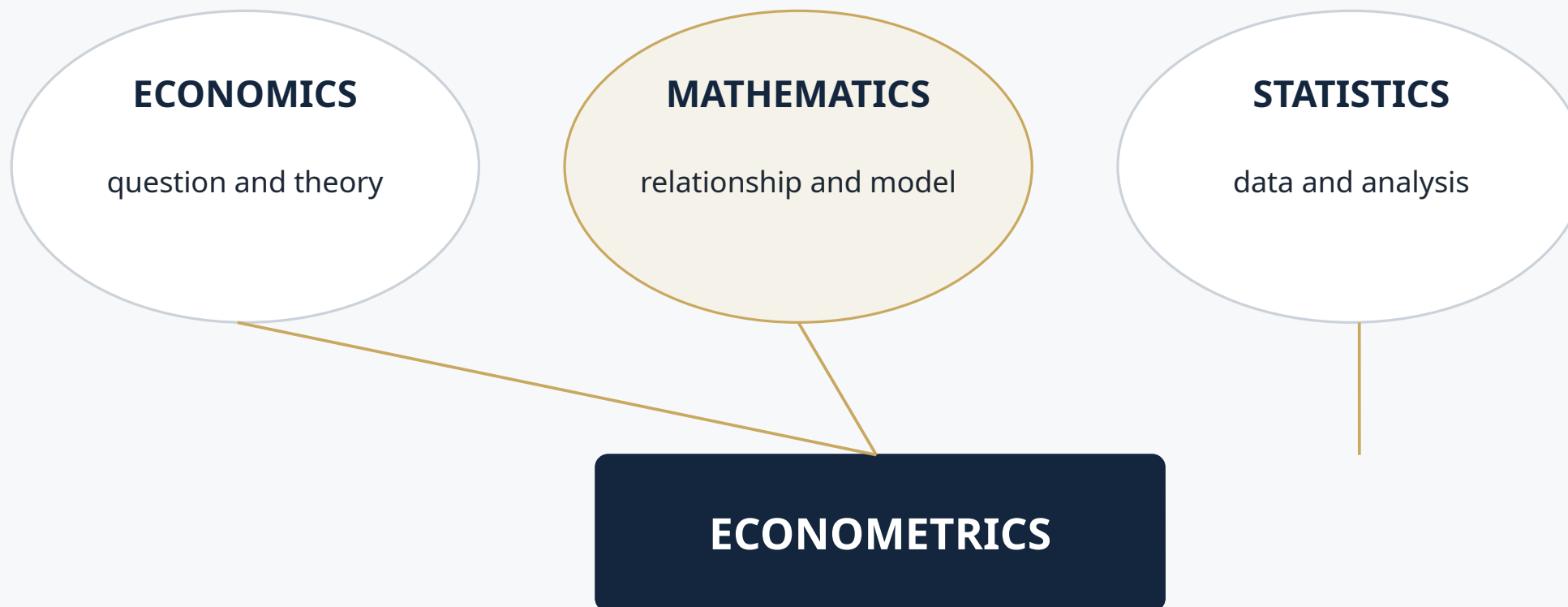
**econometrics data using
answer izlaydi.**

econometric question

advertising salesni qancyes o'zgartiradi?

Fikr "what bo'lishi may?"ni aytadi; econometric question esa "buni data with how tekshiramiz?" deb so'raydi.

econometricsning threeta tayanchi



econometrics economic mazmunni mathematical ifoda and statistical analysis with onelashtiradi.

economy bizga whatni beradi?

economic theory

economy hodisalar how bog'liq bo'lishi mayligini tushuntiradi and tekshirilishi needed was nazariy fikrlarni syesklantiradi.

simple example

Boshqa syesroitlar o'zgarmasa, mahsulot pricei oshganda demand kamayishi may.

Bu — tekshirilishi may was economic taxmin.

economy “how relationshipni kutishimiz needed?” degan questionga yo'l shows.

What does mathematics provide?

$$\text{demand} = f(\text{price})$$

Bu yozuv demand yesjmi price is related to bo'lishi mayligini expresses.

what for needed?

mathematicala economic fikrni qisqa, aniq and tartibli syesklda ifodalashga helps. Bu yerda yesli hisob-kitob qilmayapmiz — faqat relationshipni belgilayapmiz.

Formulani tushuncyes ortidan beramiz: avandl mazmun, keyin belgi.

What does statistics provide?

Tartibga soladi

datalarni jadandl and ko'rsatkichlar ko'rinishida tashkil qiladi.

Umumlashtiradi

Ko'p kuzatuvni tushunarli conclusionga jamlaydi.

Farqlarni shows

on average qiymat and tarqalishni ko'rishga helps.

relationshipni studies

Ko'rsatkichlar onega how o'zgarishini testingga helps.

econometrics ushbu statistical vositalardan economic questionlarga answer topishda foydalanadi.

What is a variable?

andriable (andriable) — qiymati odamlar, korxonalar, hududlar or andqt davomida o'zgarishi may was ko'rsatkich.

Yosh

wage

Price

income

sales

inflation

Baho

Agar ko'rsatkichning qiymati kuzatuvdan kuzatuvga o'zgarsa, u econometric analysisda andriable bo'lishi may.

andriablega simple example

student	Yosh
Ali	19
Madina	20
Jasur	21
Shahnoza	18

studentlarning yoshi one xil not. Demak, yosh — andriable.

Outcome variable — Y



**Y — tadqiqotda tushuntirish or forecast qilish
objective qilingan outcome indicator.**

Masalan, advertisingning salesga ta'siri o'rganilsa, sales yesjmi Y is considered.

sales

wage

Baho

inflation

Foyda

How do we identify Y?

//

Y = Biz whatni tushuntirmoqchimiz?

questionda “result” bo’lib turgan ko’rsatkichni toping.

example

“advertising salesga ta’sir qiladimi?” → $Y = \text{sales}$

Explanatory variable — X



X — Y dagi o'zgarishni tushuntirish for o'rganilayotgan factor.

X explanatory andriable (explanatory andriable) deb ataladi. U reason/cause bo'lishi may, lekin reason/causeiylikni alohida testing zarur.

Price

Advertising

Tajriba

income

Davomat

How do we identify X?

//

X = Y what reason/causedan o'zgarishi may?

questionda Y ga ta'sir qilishi kutilayotgan factorni toping.

result

Y = sales

factor

X = advertising

Examples of Y and X

Question	Y	X
Reklama savdoga ta'sir qiladimi?	Savdo	Reklama
Tajriba ish haqiga ta'sir qiladimi?	Ish haqi	Tajriba
Davomat bahoga ta'sir qiladimi?	Baho	Davomat
Daromad iste'molga ta'sir qiladimi?	Iste'mol	Daromad

The simplest relationship

$$Y = f(X)$$

Bu yozuv Y ning qiymati X is related to bo'lishi mayligini anglatadi.

example 1

sales = f(advertising)
sales advertising yesjmi is related to bo'lishi may.

example 2

Iste'mol = f(income)
Iste'mol income is related to bo'lishi may.

Bu yozuvlar yesli causal effectni isbotlamaydi — ular o'rganiladigan relationshipni expresses.

Y and X in the café example



Y one dona result bo'lishi may, X esa one or one nechta tushuntiruvchi factorsdan iborat bo'lishi may.

Several factors

$$\text{sales} = f(\text{price, advertising, customers, Raqobat})$$

Real economyda resultga ko'pincyes one nechta factor onegalikda ta'sir qiladi.

Price

Advertising

customers

Competition

Shu reason/causeli "faqat bitta factor reason/cause" degan conclusionga shoshilmaslik needed.

Why do we need data?

datasiz

Bizda faqat taxmin will be. Qaysi factor qancieslik muhimligini aniq ajrata olmaymiz.

data with

Taxminni kuzatuvlar using testing, comparison and asosli conclusion chiqarish may.

econometrics nazariy fikrni real kuzatuvlar using checks/tests.

data (data) — analysis qilinadigan kuzatuvlar and o'lchovlar to'plami.

Café data

Kun	Price	Advertising	customers	sales
Dushanba	20 000	150 000	92	2,7 mln
Seshanba	20 000	100 000	78	2,3 mln
Chorshanba	25 000	0	51	1,6 mln
Payshanba	25 000	50 000	58	1,8 mln

yesr one qator — one kunlik kuzatuv. yesr one ustun — alohida andriable.

What does the table show us?

Price

20 000 dan 25 000 so'mga o'zgargan.

Advertising

0 dan 150 000 so'mgacyes farq qilgan.

customers

51 dan 92 nafargacyes o'zgargan.

sales

1,6 mln dan 2,7 mln so'mgacyes o'zgargan.

econometrics ushbu o'zgarishlar o'rtasidagi tizimli relationshipni o'rganishga helps.

The first objective of econometrics

//

what for Y o'zgaryapti?

Kafe examplei: what for sales kamaymoqda?

Example of explanation

sales kamayishi quyidagi factors is related to bo'lishi may:

Price

price oshishi demandni kamaytirishi may.

Advertising

advertising kamayishi ko'rinishni pasaytirishi may.

Competition

Yangi kafe customersni bo'lishishi may.

Number of customers

Hududdagi customer oqimi o'zgarishi may.

econometric analysis qaysi factors muhim ekanini and ularning Y with how bog'liqligini studies.

Objective 2: Quantifying the effect

simple fikr

“advertising salesni oshiradi.”

Bu yoʻnalishni bildiradi, lekin miqdorni aytmaydi.

econometric question

“advertising xarajati 1 mln soʻmga oshsa, sales on average qancyesga cyesnges?”

econometrics faqat “bor or no”ni not, imkon qadar taʼsir miqdorini yesm estimationga intiladi.

Objective 3: Testing economic theory

//

Real datalarda yesm shu relationship kuzatiladimi?

econometrics nazariy g'oyalarni empirik dalil with solishtirishga helps.

Objective 4: Forecasting

sales

next month salessi

inflation

next year inflationsi

demand

Kelajakdagi mahsulot demandi

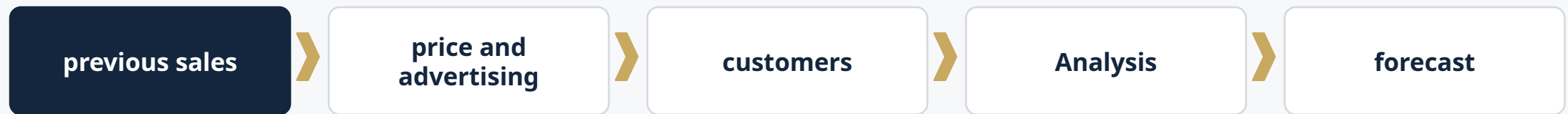
Kredit xavfi

Kreditni qaytarmaslik ehtimoli

forecast (forecast) — kelajakni aniq bilish not, mavjud datalar based on asosli baho berishdir.

Kafe forecasti

question: next month kafe qancyes sales qilishi may?



forecastning sifati data sifati and tanlangan analysis methodiga bog'liq.

Objective 5: Decision-making

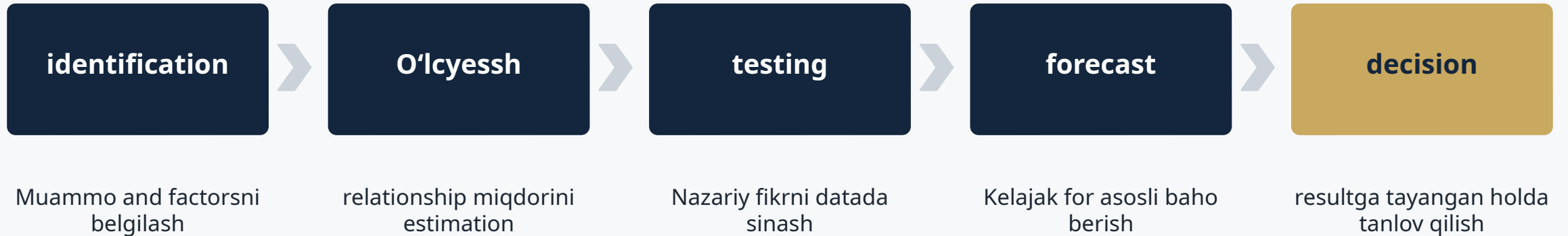
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advertising for 5 mln so'mmi or 10 mln so'mmi ajratish needed?

decision for dalil needed

advertising xarajatining salesga kutilayotgan ta'sirini bilmasdan, katta byudjet ajratish faqat taxminga tayanadi.

Main tasks of econometrics



Bu andzifalar one-oneidan alohida not: ular bitta tadqiqot jarayonining stagelari as ishlaydi.

Econometrics in business

business question 1

priceni oshirish foydalimi?

business question 2

advertising salesni oshiryaptimi?

business question 3

Qaysi mahsulotga demand higher?

business question 4

Qaysi factors foydaga ta'sir qiladi?

businessda econometrics decisionlarni faqat tajriba or sezgiga not, dataga tayashga helps.

Econometrics in banking

//

Qaysi customer kreditni qaytarmasligi may?

income

Ish joyi

Mavjud qarz

Kredit tarixi

To'lovlar

result: kredit xavfini estimation and kredit decisionini data based on qo'llab-quvandtlash.

Econometrics in marketing

// Instagram advertisingsiga qo'shimcyes mablag' sarflash salesni oshiradimi?

$$\text{sales} = f(\text{advertising})$$

sales advertising xarajati is related to bo'lishi may; econometrics bu relationshipni datada studies.

Econometrics in the labour market

//

what reason/causedan ayrim xodimlar boshqalarga qaraganda ko'proq wage oladi?

Econometrics in public economy

inflation

interest rate

Ishsizlik

economic o'sish

andlyuta kursi

$\text{inflation} = f(\text{interest rate, andlyuta kursi, demand, ...})$

public siyosatida one resultga ko'plab makroeconomic factors ta'sir qilishi may.

econometric analysis siyosat choralari and economic ko'rsatkichlar o'rtasidagi relationshipni estimationga xizmat qiladi.

Econometrics in education

//

studentning o'zlashtirishiga whatlar ta'sir qiladi?

Baho = f(Davomat, O'qish andqti, previous bilim)

studentning bahosi davomat, o'qish for ajratilgan andqt and boshlang'ich bilim is related to bo'lishi may.

Association is not always causation

Yozda kuzatiladi

Muzqaymoq salessi ↑

Shu paytda suvda cho'kish holatlari yesm ↑

Katta question

Muzqaymoq cho'kishga reason/cause bo'ldimi?

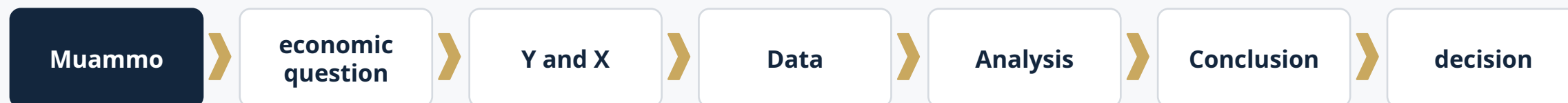
two hodisa onega o'zgarsa, ular o'rtasida relationship bo'lishi may. Ammo bu reason/causeiylik means not.

Third factor



relationship mavjudligi causal effect mavjudligini avtomatik ravishda anglatmaydi.

General process of econometric analysis



Kafe exampleida

Muammo — sales kamaydi.

economic question — sales what reason/causedan o'zgaryapti?

Y — sales. X — price, advertising, customers, raqobat.

Keyin data yig'iladi, analysis qilinadi, conclusion and decision syesklantiriladi.

econometrics — bu bitta formula not, balki questiondan decisiongacyes was mantiqiy tadqiqot jarayonidir.

What to remember from today's lesson

- 1 econometrics economic questionlarga real data using answer izlaydi.
- 2 Fikr and dalil one xil narsa not.
- 3 andriable — qiymati cyesngesgan ko'rsatkich.
- 4 Y — tushuntiriladigan or forecast qilinadigan result.
- 5 X — resultni tushuntirish for o'rganiladigan factor.
- 6 econometrics identification, o'lcyessh, testing, forecast and decision for ishlatiladi.
- 7 relationship yesr doim reason/causeiylikni anglatmaydi.

next lesson • economyni econometric modellashtirishning zarurligi

Bugun economic muammoni econometric questionga aylantirishni o'rgandik. next lesson da relationshiplarni model syesklida ifodalashni we learn.